

Experiment 4.1: FIR Low Pass Filter using Rectangular Window

Aim: To design a low-pass FIR filter using rectangular window.

MATLAB Code:

```
clc;
clear;
close all;

%% Save Command Window Output
diary('Exp4_1_Output.txt');

%% Input Parameters
fp = 1000;      % Passband cutoff frequency (Hz)
fs = 4000;      % Sampling frequency (Hz)
N = 50;         % Filter order

wc = fp/(fs/2); % Normalized cutoff frequency

%% FIR Low Pass Filter using Rectangular Window
b_rect = fir1(N, wc, 'low', rectwin(N+1));

disp('--- Rectangular Window Coefficients ---');
disp(b_rect);

%% Frequency Response
figure(1);
freqz(b_rect,1);
title('Low-pass FIR using Rectangular Window');

%% Stop Saving Command Window
diary off;

%% Save Figure (300 DPI)
exportgraphics(figure(1), 'Exp4_1_Rectangular_Window.png', 'Resolution',300);

%% Completion Message
disp(' ');
disp('=====');
disp('Experiment 4.1 Completed Successfully');
disp('Files Saved:');
disp('Exp4_1_Output.txt');
disp('Exp4_1_Rectangular_Window.png');
disp('=====');
```

Output:

```
--- Rectangular Window Coefficients ---
0.0126
0
-0.0137
0
0.0150
0
-0.0166
0
0.0185
0
-0.0210
0
0.0242
```

0
-0.0286
0
0.0349
0
-0.0449
0
0.0629
0
-0.1048
0
0.3145
0.4940
0.3145
0
-0.1048
0
0.0629
0
-0.0449
0
0.0349
0
-0.0286
0
0.0242
0
-0.0210
0
0.0185
0
-0.0166
0
0.0150
0
-0.0137
0
0.0126

Result:

